

Lecture 08 — Poisson Process, Jump Diffusion, and Affine Jump Diffusion

Computational Finance (WI4151), TU Delft — exam notes

Scope. This lecture builds the machinery for asset-price models with discontinuous paths. The logical chain is: (i) the Poisson process as a counting process; (ii) attaching random jump sizes to it; (iii) the jump-diffusion SDE for the stock and the drift correction forced by the risk-neutral martingale condition; (iv) the characteristic function (cf) of the resulting log-price, which is what Fourier/COS methods consume; (v) the affine jump-diffusion (AJD) framework (Duffie–Pan–Singleton 2000) that generalises Heston, Merton and their union, and reduces the cf to a system of Riccati ODEs. The recurring punchline of the course is on slide 32: *the COS method works as long as the characteristic function is known*. Everything here exists so that we can write down that cf. Cross-reference [notes/Lecture06-notes.pdf](#) for how the cf actually enters the European COS formula.

1 Why jumps: the empirical motivation

The Black–Scholes/GBM model has continuous paths and Gaussian log-returns. Empirically (slides 3–4), equity index returns show (a) sudden large moves — the SPX fell sharply at the onset of COVID-19 in March 2020 — and (b) *fat tails*: the histogram of daily SPX returns has far more mass in the extremes than the fitted normal, and the Q–Q plot bends away from the diagonal at both ends. Fat tails are quantified by the standardised third and fourth moments,

$$\text{skewness} = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^3 \right], \quad \text{kurtosis} = \mathbb{E} \left[\left(\frac{X - \mu}{\sigma} \right)^4 \right],$$

which for the normal are 0 and 3; observed returns have negative skew and excess kurtosis (> 3). Diffusion alone cannot generate this at short horizons (a small-variance Gaussian is thin-tailed), so we add a jump component. Jumps are also the natural way to model the unpredictability of crashes: they arrive at random times that are not foreseeable from past information.

2 The Poisson random variable and the Poisson process

2.1 Poisson distribution

A discrete random variable X is Poisson with parameter $\lambda > 0$ if

$$f(k) = \mathbb{P}(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad k \in \mathbb{N}. \quad (1)$$

Its mean is λ , computed directly:

$$\mathbb{E}[X] = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!} = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!} = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} e^{\lambda} = \lambda,$$

using $\sum_{j \geq 0} \lambda^j / j! = e^{\lambda}$. A short companion computation of $\mathbb{E}[X(X-1)] = \lambda^2$ gives $\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda$. So for the Poisson law, *mean = variance* = λ ; this is a useful sanity check and an examiner favourite.

2.2 From counts to a process

Fix an interval length t and model the number of events (jumps) in it as Poisson with parameter λt :

$$\mathbb{P}(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}, \quad k \in \mathbb{N}.$$

The **(homogeneous) Poisson process** N_t with intensity λ is the counting process satisfying

- (i) $N_0 = 0$;
- (ii) independent and stationary increments;
- (iii) $N_t - N_s \sim \text{Poisson}(\lambda(t-s))$ for $s < t$.

Equivalently we write $N_t = \int_0^t dN_s = \sum_{\tau_k \leq t} dN(\tau_k)$, where $\{\tau_k\}$ are the (random) jump times. The increments being stationary means the law of $N_{t+\Delta t} - N_t$ depends only on Δt , not on t ; combined with independence this pins the process down up to λ .

2.3 Compensated Poisson process and the martingale property

N_t has positive drift ($\mathbb{E}[N_t] = \lambda t$), so it is not a martingale. Subtracting its mean gives the **compensated Poisson process**

$$\tilde{N}_t := \int_0^t dN_s - \lambda t = N_t - \lambda t,$$

which *is* a martingale. The verification (slide 9) checks the three ingredients of a martingale: $N_{t+\Delta t} - N_t$ is independent of $\mathcal{F}_t$ and stationary; $\mathbb{E}[N_{t+\Delta t} - N_t] = \lambda \Delta t$, so

$$\mathbb{E}[\tilde{N}_{t+\Delta t} | \mathcal{F}_t] = \tilde{N}_t + \mathbb{E}[N_{t+\Delta t} - N_t] - \lambda \Delta t = \tilde{N}_t + \lambda \Delta t - \lambda \Delta t = \tilde{N}_t;$$

and integrability, $\mathbb{E}[|\int_0^t dN_s - \lambda t|] < \mathbb{E}[N_t] + \lambda t = 2\lambda t < \infty$. The constant λt is the *compensator*: it is the predictable, finite-variation part that must be removed to leave a martingale. This compensator idea returns immediately when we price under $\mathbb{Q}$: the risk-neutral drift correction is exactly the compensator of the jump term.

2.4 Time-dependent and stochastic intensity

If the jump probability varies in time, let $\lambda = \lambda(t)$ and define the integrated intensity $\Lambda(a, b) = \int_a^b \lambda(s) ds$. Then for $N_{(a,b]} \equiv N_b - N_a$,

$$\mathbb{P}(N_{(a,b]} = k) = \frac{\Lambda(a, b)^k e^{-\Lambda(a, b)}}{k!},$$

and N_t is an **inhomogeneous Poisson process**: $N_0 = 0$, independent increments, $\mathbb{P}(N_{t+h} - N_t = 1) = \lambda(t)h + O(h^2)$ and $\mathbb{P}(N_{t+h} - N_t \geq 2) = O(h^2)$. The $O(h^2)$ rule (no two jumps in an infinitesimal interval) is what later justifies $dN_t \in \{0, 1\}$. Again $\int_0^t dN_s - \int_0^t \lambda(s) ds$ is a martingale.

If $\lambda(t)$ is itself a stochastic process (adapted to the same filtration, with $\int_0^T \lambda(t) dt < \infty$ a.s.), N_t is a **Cox / doubly stochastic Poisson process**. Conditional on a realised path of $\lambda(\cdot)$ it is just an inhomogeneous Poisson process, so

$$\mathbb{P}(N_{(a,b]} = k | \mathcal{F}_a) = \mathbb{E} \left[\frac{\Lambda(a, b)^k e^{-\Lambda(a, b)}}{k!} \middle| \mathcal{F}_a \right],$$

and the compensated process $\int_0^t dN_s - \int_0^t \lambda(s) ds$ remains a martingale. The Cox construction is the natural way to let intensity respond to the state (e.g. intensity rising with variance), and it is precisely the AJD slot $\lambda(x) = l_0 + l_1 \cdot x$ later.

Remark 1 (Paths are càdlàg). *Poisson paths are càdlàg (right-continuous with left limits): at a jump time t the value is N_t , while $N_{t-} = \lim_{s \uparrow t} N_s$ is the pre-jump value. The jump is not predictable from $\mathcal{F}_{t-}$; this is the mathematical statement that crashes cannot be anticipated. Throughout, “ t^- ” denotes the left limit and matters once we attach state-dependent jump sizes.*

3 The compound Poisson process and its characteristic function

Attach to each jump an i.i.d. random size. The **compound Poisson process** is

$$Q_t = \sum_{k=1}^{N_t} Y_k,$$

with $\{Y_k\}$ i.i.d. (law ν , cf $\hat{\nu}(u) = \mathbb{E}[e^{iuY}]$), independent of the Poisson count N_t . The cf of Q_t is the single most reusable computation in this lecture; it is what makes jumps tractable for Fourier pricing.

3.1 Derivation by conditioning (the characteristic exponent)

Condition on the number of jumps and use independence:

$$\mathbb{E}[e^{iuQ_t}] = \sum_{n=0}^{\infty} \mathbb{P}(N_t = n) \mathbb{E}[e^{iu \sum_{k=1}^n Y_k}] = \sum_{n=0}^{\infty} \frac{(\lambda t)^n e^{-\lambda t}}{n!} (\hat{\nu}(u))^n,$$

because the Y_k are i.i.d. so $\mathbb{E}[e^{iu \sum_{k=1}^n Y_k}] = \hat{\nu}(u)^n$. Pull out $e^{-\lambda t}$ and recognise the exponential series:

$$\mathbb{E}[e^{iuQ_t}] = e^{-\lambda t} \sum_{n=0}^{\infty} \frac{(\lambda t \hat{\nu}(u))^n}{n!} = e^{-\lambda t} e^{\lambda t \hat{\nu}(u)} = \exp(\lambda t (\hat{\nu}(u) - 1)).$$

Hence

$$\varphi_{Q_t}(u) = \exp(t \lambda (\hat{\nu}(u) - 1)) = \exp\left(t \lambda \int_{\mathbb{R}} (e^{iuy} - 1) d\nu(y)\right). \quad (2)$$

The exponent is *linear in t* , so the **characteristic exponent** is $\psi(u) = \lambda(\hat{\nu}(u) - 1)$. This is the simplest non-trivial instance of the **Lévy–Khintchine** representation: a pure-jump Lévy process has cf $\exp(t\psi(u))$ with $\psi(u) = \int_{\mathbb{R}} (e^{iuy} - 1) \Pi(dy)$ and Lévy measure $\Pi(dy) = \lambda \nu(dy)$ (finite mass λ , i.e. finite activity). The “ -1 ” inside is the compensation that makes $\psi(0) = 0$ (so $\varphi_{Q_t}(0) = 1$, as any cf must satisfy).

3.2 Adding the diffusion: cf of a Lévy jump-diffusion log-price

For a process $X_t = \gamma t + \sigma W_t + Q_t$ (drift + Brownian + compound Poisson, all independent), cfs multiply, so

$$\varphi_{X_t}(u) = \exp\left(t \left[iu\gamma - \frac{1}{2}\sigma^2 u^2 + \lambda(\hat{\nu}(u) - 1) \right]\right). \quad (3)$$

This is the full Lévy–Khintchine exponent (Gaussian part $-\frac{1}{2}\sigma^2 u^2$, drift $iu\gamma$, jump part $\lambda(\hat{\nu}(u) - 1)$). For pricing we will fix γ by the martingale condition of §5, but the *shape* (3) is model-independent: choose the jump law ν and you have the cf. This is the bridge to COS.

4 The jump-diffusion SDE and its solution

4.1 Random jump sizes and Itô for jumps

Start from $dY_t = J_t dN_t$: a jump of size J_t occurs whenever $dN_t = 1$ (and $dN_t \in \{0, 1\}$ because two simultaneous jumps have probability $O(dt^2)$). J_t is taken left-continuous. For a general

jump-diffusion $dY_t = \mu_t dt + \sigma_t dW_t + J_t dN_t$, Itô's lemma must add the jump contribution *exactly* (not via a Taylor expansion):

$$dg(Y_t, t) = \frac{\partial g}{\partial t} dt + \frac{\partial g}{\partial Y} (\mu_t dt + \sigma_t dW_t) + \frac{1}{2} \frac{\partial^2 g}{\partial Y^2} \sigma_t^2 dt + [g(Y_{t-} + J_t, t) - g(Y_{t-}, t)] dN_t. \quad (4)$$

The first three terms act on the *continuous part* $Y_t^c = Y_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s$ (jumps excluded); the bracket is the *finite* change in g across a jump, not its differential. There is no $\frac{1}{2}g''J^2$ term — a jump is a finite move, so we use the exact difference. The product rule generalises with a covariation term $d[Y, Z]_t = \sigma_t^Y \sigma_t^Z \rho dt + \sum_{s \leq t} J_s^Y J_s^Z$ (continuous co-quadratic-variation plus simultaneous jumps); for $Y = Z$ this is the quadratic variation $[Y, Y]_t = \int_0^t (\sigma_s^Y)^2 ds + \sum_{s \leq t} (J_s^Y)^2$.

4.2 The stock SDE and its log

The Merton-type stock dynamics are

$$dS_t = \mu S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t, \quad (5)$$

with J_t a continuous random variable independent of S_t and N_t . At a jump, $\Delta S_t = S_t - S_{t-} = J_t S_{t-}$, so $S_t = S_{t-}(1 + J_t)$: the price is multiplied by $(1 + J_t)$. (For $S_t > 0$ we need $J_t > -1$.) Apply (4) with $g = \ln$:

$$d \ln S_t = \frac{1}{S_{t-}} (\mu S_{t-} dt + \sigma S_{t-} dW_t) - \frac{1}{2 S_{t-}^2} \sigma^2 S_{t-}^2 dt + [\ln S_t - \ln S_{t-}] dN_t,$$

and since $\ln S_t - \ln S_{t-} = \ln(1 + J_t)$ at a jump,

$$d \ln S_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t + \ln(1 + J_t) dN_t. \quad (6)$$

So the log-price is exactly the Lévy jump-diffusion of (3) with jump variable $\ln(1 + J_t)$. Integrating, $\ln S_T = \ln S_0 + (\mu - \frac{1}{2} \sigma^2)T + \sigma W_T + \sum_{k=1}^{N_T} \ln(1 + J_k)$: a Brownian log plus a compound-Poisson sum of i.i.d. log-jumps.

5 The compensator and the risk-neutral drift correction

The pricing requirement is that the discounted stock $S_t/M_t = e^{-rt} S_t$ is a $\mathbb{Q}$ -martingale. Apply the product rule to $e^{-rt} S_t$ (the cross term $de^{-rt} dS_t = 0$ since e^{-rt} has finite variation):

$$d(S_t/M_t) = e^{-rt} [-r S_{t-} dt + dS_t] = e^{-rt} [(\mu - r) S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t].$$

The dW_t term is already a martingale. For the rest, take $\mathbb{E}^\mathbb{Q}$ and use $\mathbb{E}[dN_t] = \lambda dt$ together with the independence of J_t :

$$\mathbb{E}^\mathbb{Q}[d(S_t/M_t)] = e^{-rt} S_{t-} \mathbb{E}^\mathbb{Q}[(\mu - r) dt + J_t dN_t] = e^{-rt} S_{t-} [(\mu - r) + \lambda \mathbb{E}^\mathbb{Q}[J_t]] dt.$$

Setting this to zero gives the **risk-neutral drift correction**

$$\boxed{\mu = r - \lambda \mathbb{E}^\mathbb{Q}[J_t]}. \quad (7)$$

The term $\lambda \mathbb{E}^\mathbb{Q}[J_t]$ is the *jump compensator*: it is exactly $\mathbb{E}[J_t dN_t]/dt$, the mean rate at which jumps push the price up, and it must be subtracted from the drift so the jumps do not bias the expected return away from r . Equivalently, $J_t dN_t - \lambda \mathbb{E}^\mathbb{Q}[J_t] dt$ is the compensated (martingale) jump term, mirroring § 2.3. The $\mathbb{Q}$ stock dynamics are

$$dS_t = (r - \lambda \mathbb{E}^\mathbb{Q}[J_t]) S_{t-} dt + \sigma S_{t-} dW_t + J_t S_{t-} dN_t.$$

Substituting (7) into (6),

$$d \ln S_t = \left(r - \lambda \mathbb{E}^\mathbb{Q}[J_t] - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t + \ln(1 + J_t) dN_t. \quad (8)$$

Remark 2 (Compensate the right thing). *A standard trap: the drift correction uses $\mathbb{E}[J_t]$ (the arithmetic jump in $S/S_{t-} - 1$), not $\mathbb{E}[\ln(1 + J_t)]$. In the Merton parametrisation below one usually models $\ln(1 + J) = \xi \sim \mathcal{N}(\mu_J, \sigma_J^2)$, so $1 + J = e^\xi$ and $\mathbb{E}[J] = \mathbb{E}[e^\xi] - 1 = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1$. The compensator is $\lambda(e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1)$. Mixing the two is the classic sign/exponent error.*

5.1 PIDE for the option value

Requiring V_t/M_t a martingale and repeating the Itô-for-jumps expansion on $e^{-rt}V(S_t, t)$ gives, after taking $\mathbb{E}^\mathbb{Q}$ and using independence of J , the **partial integro-differential equation (PIDE)**

$$\frac{\partial V}{\partial t} + (r - \lambda \mathbb{E}[J])S \frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} - rV + \lambda \mathbb{E}^\mathbb{Q}[V(S(1 + J), t) - V(S, t)] = 0.$$

The last term — an integral against the jump law — is the non-local piece that distinguishes a PIDE from the BS PDE. Numerically it couples all S , which is exactly why a transform method (read off the cf once, integrate against the payoff) is so much cleaner than solving the PIDE on a grid.

6 Two popular jump-size models and their cf

With $\xi = \ln(1 + J)$ the log-jump, the only model-specific input to (3) is $\widehat{\nu}(u) = \mathbb{E}[e^{iu\xi}]$.

6.1 Merton (1976): lognormal jumps

Log-jumps are Gaussian, $\xi \sim \mathcal{N}(\mu_J, \sigma_J^2)$, with density

$$\nu_J(x) = \frac{1}{\sigma_J \sqrt{2\pi}} \exp\left(-\frac{(x - \mu_J)^2}{2\sigma_J^2}\right), \quad \widehat{\nu}(u) = \exp\left(iu\mu_J - \frac{1}{2}\sigma_J^2 u^2\right).$$

Plugging into (3) with the risk-neutral drift $\gamma = r - \frac{1}{2}\sigma^2 - \lambda \mathbb{E}[J]$ and $\mathbb{E}[J] = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1$ gives the **Merton cf of the log-price** $\ln S_T$ (writing $x = \ln S_0$, $\tau = T$):

$$\varphi_{\ln S_T}(u) = \exp\left(iu\left(x + \left(r - \frac{1}{2}\sigma^2 - \lambda\kappa\right)\tau\right) - \frac{1}{2}\sigma^2 u^2 \tau + \lambda\tau\left(e^{iu\mu_J - \frac{1}{2}\sigma_J^2 u^2} - 1\right)\right), \quad (9)$$

where $\kappa = \mathbb{E}[J] = e^{\mu_J + \frac{1}{2}\sigma_J^2} - 1$. The first two terms are the GBM cf; the last is the compound-Poisson exponent (2). Effects on implied volatility (slide 31): σ_J controls the smile *curvature*, λ the overall *level*, and μ_J the *slope/skew* (negative $\mu_J \Rightarrow$ downward skew, the equity case).

6.2 Kou (2002): double-exponential jumps

Log-jumps follow an asymmetric double-exponential law,

$$\nu_X(x) = p\eta_1 e^{-\eta_1 x} \mathbf{1}_{\{x \geq 0\}} + q\eta_2 e^{\eta_2 x} \mathbf{1}_{\{x < 0\}}, \quad p + q = 1, \quad \eta_1 > 1, \quad \eta_2 > 0.$$

Up-jumps are $\text{Exp}(\eta_1)$ with probability p , down-jumps are $-\text{Exp}(\eta_2)$ with probability q . Its transform (Laplace/cf) is

$$\widehat{\nu}(u) = \mathbb{E}[e^{iuX}] = p \frac{\eta_1}{\eta_1 - iu} + q \frac{\eta_2}{\eta_2 + iu},$$

obtained by integrating each exponential tail. The condition $\eta_1 > 1$ guarantees $\mathbb{E}[e^X] < \infty$, i.e. the jump multiplier $1 + J = e^X$ has finite mean so the compensator $\kappa = \mathbb{E}[J] = p \frac{\eta_1}{\eta_1 - 1} + q \frac{\eta_2}{\eta_2 + 1} - 1$ is finite. The Kou cf of the log-price is again (3) with this $\widehat{\nu}$ and $\gamma = r - \frac{1}{2}\sigma^2 - \lambda\kappa$:

$$\varphi_{\ln S_T}(u) = \exp\left(iu\left(x + \left(r - \frac{1}{2}\sigma^2 - \lambda\kappa\right)\tau\right) - \frac{1}{2}\sigma^2 u^2 \tau + \lambda\tau\left(\widehat{\nu}(u) - 1\right)\right).$$

Remark 3 (Merton vs. Kou). *Both add finite-activity jumps to GBM and both yield closed-form cfs. Merton: symmetric-tailed Gaussian jumps, smooth, but the cf decays like e^{-cu^2} (rapidly). Kou: exponential tails $\Rightarrow$ heavier, asymmetric, and the cf decays only like $1/u^2$ (rationally); Kou jumps admit semi-analytic results for barriers/first-passage because of the memoryless exponential, which Merton does not. For COS the practical difference is the tail decay of φ : slower decay (Kou) means more cosine terms N for the same accuracy, and the cumulant-based range $[a, b]$ is wider because exponential jumps fatten the tails.*

7 How the cf enters COS (the crucial connection)

Recall (Lecture 06) the European COS price is

$$v(x_0, t_0) \approx e^{-r\tau} \sum_{k=0}^{N-1} {}' \operatorname{Re} \left\{ \varphi_{X_T} \left(\frac{k\pi}{b-a} \right) e^{-ik\pi \frac{a}{b-a}} \right\} V_k,$$

where the prime halves the $k = 0$ term, V_k are the (closed-form) cosine coefficients of the payoff on $[a, b]$, and φ_{X_T} is the cf of the log-price (with the spot dependence carried as $\varphi_{X_T}(u) = \varphi(u)e^{iux_0}$). The entire model enters through φ *evaluated at the grid* $u_k = k\pi/(b-a)$. Lecture 08 supplies φ in closed form for Merton (9) and Kou, exactly the slot COS needs — nothing else about the jump structure is required. This is the content of slide 32: COS is applicable as long as the cf is known. Two practical riders, both exam-relevant:

1. **Cumulants for $[a, b]$.** The truncation range is set from cumulants c_1, c_2, c_4 of $X_T = \ln S_T$, $[a, b] = [c_1 \pm L\sqrt{c_2 + \sqrt{c_4}}]$. For a jump-diffusion these get explicit jump contributions; e.g. for Merton, $c_1 = (r - \frac{1}{2}\sigma^2 - \lambda\kappa)\tau + \lambda\tau\mu_J$, $c_2 = \sigma^2\tau + \lambda\tau(\mu_J^2 + \sigma_J^2)$, $c_4 = \lambda\tau(\mu_J^4 + 6\mu_J^2\sigma_J^2 + 3\sigma_J^4)$. The pure jump term in c_4 is what makes the density fat-tailed and forces a wider $[a, b]$ than the diffusion alone would. *This is precisely the short-maturity trap:* at small τ the diffusion $c_2 = \sigma^2\tau$ shrinks, but jumps can dominate; if $[a, b]$ is built from c_2 alone (forgetting the jump kurtosis c_4 , or taking L too small) the range is too narrow, the truncation error floor stays high, and the price is wrong no matter how many terms N you take. Include c_4 and use a generous L .
2. **cf decay sets N .** COS series error is controlled by the tail decay of $\varphi(u_k)$. Merton decays super-exponentially (few terms); Kou only polynomially (more terms). Always evaluate φ on the principal branch to avoid the kind of branch-cut error that plagues the Heston cf.

8 Affine processes and the AJD framework

The Heston model and the Merton/Kou jump models are both special cases of one structure: **affine jump diffusions**. A Markov process is *affine* if its conditional cf is exponential-affine in the current state,

$$\mathbb{E}[e^{u \cdot X_T} \mid X_t = x] = e^{\alpha(u, t, T) + \beta(u, t, T) \cdot x}.$$

This is the structural reason the cf is available in (semi-)closed form: the x -dependence collapses to a scalar α and a vector β solving ODEs.

8.1 AJD dynamics (Duffie–Pan–Singleton, 2000)

An n -dimensional AJD is

$$dX_t = \mu(X_t) dt + \sigma(X_t) dW_t + dZ_t,$$

where every coefficient is affine in the state x :

- drift $\mu(x) = K_0 + K_1 x$, with $K_0 \in \mathbb{R}^n$, $K_1 \in \mathbb{R}^{n \times n}$;
- diffusion $[\sigma(x)\sigma(x)^\top]_{ij} = (H_0)_{ij} + (H_1)_{ij} \cdot x$, with $H_0 \in \mathbb{R}^{n \times n}$, $H_1 \in \mathbb{R}^{n \times n \times n}$;

- jumps $dZ_t = J dN_t$ with state-dependent intensity $\lambda(x) = l_0 + l_1 \cdot x$, jump sizes $J \sim \nu$ independent of X_t ;
- short rate $R(x) = \rho_0 + \rho_1 \cdot x$ (also affine), so discounting stays in the family.

The jump law enters only through its “jump transform”

$$\theta(c) = \int_{\mathbb{R}^n} e^{c \cdot z} d\nu(z),$$

i.e. the same $\hat{\nu}$ as in §3 (here written with a general complex argument $c = \beta(t)$).

8.2 The cf as the solution of Riccati ODEs

The key AJD result: the discounted transform

$$\varphi^X(u, x, t, T) = \mathbb{E} \left[\exp \left(- \int_t^T R(X_s) ds \right) e^{u \cdot X_T} \mid \mathcal{F}_t \right] = e^{\alpha(u, t, T) + \beta(u, t, T) \cdot x},$$

where (α, β) solve the complex ODEs (boundary conditions $\beta(u, T, T) = u$, $\alpha(u, T, T) = 0$)

$$\dot{\beta} = \rho_1 - K_1^\top \beta - \frac{1}{2} \beta^\top H_1 \beta - l_1 (\theta(\beta) - 1), \quad \dot{\alpha} = \rho_0 - K_0 \cdot \beta - \frac{1}{2} \beta^\top H_0 \beta - l_0 (\theta(\beta) - 1).$$

The β equation is a *Riccati* ODE (quadratic in β via the H_1 term); α is then a quadrature. The jump contribution is the same $(\theta(\beta) - 1)$ compensator-type term that appeared in (2) — the unifying observation of the whole lecture: *jumps enter every formula as $\theta(\cdot) - 1$.*

8.3 Proof sketch (why the ODEs)

Set $\varphi_t = \exp(-\int_0^t R ds) e^{\alpha(t) + \beta(t) \cdot X_t}$. By construction φ_t should be a martingale (then $\varphi_t = \mathbb{E}_t[\varphi_T]$ gives the claimed transform). Apply Itô-for-jumps. The jump term’s expected rate is

$$\mathbb{E}_t[(\varphi_t(X_{t-} + J) - \varphi_t(X_{t-})) dN_t] = \int_{\mathbb{R}^n} \varphi_{t-} (e^{\beta(t) \cdot y} - 1) d\nu(y) \lambda(X_t) dt = \varphi_t (\theta(\beta(t)) - 1) \lambda(X_t) dt,$$

which is exactly where $\theta(\beta) - 1$ comes from. Collecting the dt drift and using $\partial_t \varphi = (-R + \dot{\alpha} + \dot{\beta} \cdot X) \varphi$, $\partial_X \varphi = \beta \varphi$, $\partial_{XX} \varphi = \beta \beta^\top \varphi$, the martingale (zero-drift) condition reads

$$[-R(X) + \dot{\alpha} + \dot{\beta} \cdot X] + \mu(X) \beta + \frac{1}{2} \beta \sigma \sigma^\top \beta^\top + (\theta(\beta) - 1) \lambda(X) = 0.$$

Substituting the affine forms of $R, \mu, \sigma \sigma^\top, \lambda$ and matching the X -independent part and the coefficient of X separately (since it must hold for all X_t) yields precisely the two ODEs above.

8.4 Sanity check: Black–Scholes as an AJD

For GBM there are no jumps ($l_0 = l_1 = 0$). With $X = \ln S$, $\tau = T - t$, the pricing PDE in log-variable is $-\partial_\tau U + (r - \frac{1}{2} \sigma^2) \partial_X U + \frac{1}{2} \sigma^2 \partial_{XX} U - rU = 0$. Probing $U = \exp(\alpha(u, \tau) + \beta(u, \tau) X)$ and using $\partial_X U = \beta U$, $\partial_{XX} U = \beta^2 U$, $\partial_\tau U = (\dot{\alpha} + X \dot{\beta}) U$ gives

$$-(\dot{\alpha} + X \dot{\beta}) + (r - \frac{1}{2} \sigma^2) \beta + \frac{1}{2} \sigma^2 \beta^2 - r = 0 \Rightarrow \dot{\beta} = 0, \quad \dot{\alpha} = (r - \frac{1}{2} \sigma^2) \beta + \frac{1}{2} \sigma^2 \beta^2 - r.$$

With $\beta(u, 0) = iu$ (so $\beta \equiv iu$) and $\alpha(u, 0) = 0$,

$$\alpha(u, \tau) = (r - \frac{1}{2} \sigma^2) iu \tau - \frac{1}{2} \sigma^2 u^2 \tau - r\tau,$$

recovering the GBM log-price cf $\varphi(u, \tau) = \exp((r - \frac{1}{2} \sigma^2) iu \tau - \frac{1}{2} \sigma^2 u^2 \tau - r\tau + iu X)$. This is the same closed form COS uses for Black–Scholes, now derived as the degenerate AJD — confirming the framework specialises correctly. Adding the jump terms $l_0(\theta(iu) - 1)$ reproduces (9).

9 Where this sits in the course

- **Heston (Lecture 07)** is the AJD with state $X = (\ln S, v)$, affine variance v entering both the diffusion (H_1 term, $\sigma\sqrt{v}$) and as the CIR variance process; no jumps. Its cf is the (α, β) solution with a non-trivial Riccati β . Bates' model = Heston + Merton jumps is the canonical AJD that needs both halves of this lecture.
- **COS (Lectures 06/10)** is the consumer: it needs only φ , supplied here in closed form (Merton, Kou) or via the ODE solve (general AJD). The pipeline is: pick model $\rightarrow$ get φ from this lecture $\rightarrow$ set $[a, b]$ from cumulants (with jump $c_4!$) $\rightarrow$ sum the COS series.
- **The compensator/ $\theta - 1$ motif** appears at three scales: the compensated Poisson process $(-\lambda t)$, the risk-neutral drift correction $(-\lambda \mathbb{E}[J])$, and the AJD ODEs $(-l(\theta(\beta) - 1))$. They are the same idea — remove the predictable mean of the jumps.

Honesty note: what the slides do *not* contain

For completeness, the following are referenced or implied but not derived in the Lecture 08 deck, and I have filled them in from standard sources (Cont–Tankov; Duffie–Pan–Singleton; Fang–Oosterlee):

- The slides state the Merton density ν_J and Kou density ν_X but *do not write the jump-diffusion characteristic functions* (9) or the Kou cf explicitly; the conditioning derivation (2) and these cfs are supplied here.
- The phrase “Lévy–Khintchine” does not appear on the slides; I have flagged (2)–(3) as the finite-activity special case so the structure is named correctly.
- The Kou transform $\widehat{\nu}(u) = p\eta_1/(\eta_1 - iu) + q\eta_2/(\eta_2 + iu)$ and the explicit cumulants c_1, c_2, c_4 for the COS range are not on the slides; they are standard and added for the COS connection. *Verify the exact c_4 constant against your Assignment 2 code before the oral.*
- The slides cover the general n -dimensional AJD and the BS specialisation, but do *not* carry out the Heston specialisation of (α, β) ; see Lecture 07 notes for that Riccati solution.